Nicolas Toon

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Education

B.S. in Data Science, *University of California, San Diego* Sep. 2023 – Mar. 2027

- **GPA:** 3.92/4.0
- **Relevant Coursework:** Machine Learning Algorithms, Probabilistic Machine Learning, Data Visualization, Database Management, Data Structures & Algorithms, Probability/Calculus/Linear Algebra, Image Processing

Experience

Software Engineering Intern, *alwaysAI* May 2026 – Present

- Structuring computer vision applications as modular systems for scalable integration and deployment
- Implementing control logic, monitoring, and business rules with perception models for real-world performance
- Enhancing internal developer tools to accelerate development, debugging, and deployment processes

Software Lead, *Triton AI* Oct. 2025 – Present

- Training and deploying YOLO computer vision models with OpenCV/Ultralytics to detect go-karts in real time
- Building a Docker-based camera/LiDAR fusion pipeline on Raspberry Pi for deployment to NVIDIA Jetson
- Tuning and validating control system parameters to optimize go-kart pathfinding and kinematic states
- Refactoring legacy ROS2 software to remove deprecated components and improve maintainability
- Deploying a watchdog system that continuously monitors go-kart health and triggers failsafes to ensure safety

Bioinformatics Analyst Intern, *Scripps Research* Sep. 2024 – Jun. 2026

- Wrote Python functions with matplotlib and Plotly, synthesizing tabular data into time series visualizations
- Generated understandable deliverables reporting significant genetic mutations found using pandas and Dask
- Created a dashboard allowing researchers to visualize cell types and coexpressed genes in a mouse brain

UI Designer & Software Developer, *Data Science Student Society* Oct. 2025 – Jun. 2026

- Designed and created websites using Figma, React, and Tailwind for San Diego organizations and events
- Developed backend integrations including webhook-driven dashboard updates and email verification workflows

Projects

Stage Left: Typing with Parkinson's Jun. 2025

- Collaborated with two undergraduates to develop an interactive and educational web tool for the general public
- Utilized D3 to create a custom typing game to race against a model, which simulated a Parkinson's patient

Classifying Benign and Malignant Breast Tumors Jun. 2025

- Researched and compared image segmentation architectures, including U-Nets and vision transformers
- Built a full U-Net computer vision pipeline in PyTorch, from dataset curation to model training and evaluation
- Integrated a binary classifier with segmentation, achieving 90.2% recall on malignant tumor detection

Recipe Representation Mar. 2025

- Conducted exploratory data analysis and hypothesis tests to verify the significance behind online recipe tags
- Constructed a classifier with scikit-learn to make predictions about tags based on a given recipe's data

Skills

Coding Languages: Python, SQL, JavaScript/TypeScript, Java, C++, HTML/CSS

Software: Git, Docker, ROS2, Jenkins, AWS (EC2, S3), Linux/Bash

Computer Vision: OpenCV, PyTorch, Ultralytics, DepthAI, CVAT, Roboflow

Data Modeling: pandas, NumPy, SciPy, matplotlib, Plotly, Tableau, scikit-learn, Dask, D3, Excel